

All Tracks & What's Left to Ship a Real Product

Product	AI CMO for Indian founders — FastAPI + LangGraph backend, Next.js frontend, Clerk auth, Supabase data.
Now live	Clerk sign-in · automatic founder provisioning on signup (webhook) · onboarding saves the full founder profile into Supabase · backend running with resilient startup.
This doc	Every track (backend + frontend + models + scale), what's done vs left, and a realistic estimate to a deployable, real-data V1.

Backend engine
~80% — auth, data, intake,
webhook, resilient boot all live

Frontend product
~45% — auth + onboarding real;
dashboards still mock

To real-data V1
~5–7 focused weeks (excl. trained
models)

1. Where we are now

MRK18 has crossed the hardest line for an early product: real users now produce real data. A Clerk signup automatically creates a founder row in Supabase (verified live), and the onboarding form writes a gate-validated founder profile through the real backend. The backend's core — auth, tenant isolation, the token vault, the hash-chained audit log, the LangGraph approval pipeline, 204 passing tests — was already production-grade; since the June audit we've added the integration layer (Clerk auth, CORS, founder provisioning, intake) and hardened startup. What remains is finishing the security hardening, completing the integration surface, making it deployable, and — the biggest single piece — rebuilding the frontend on the real product instead of mock data.

2. The tracks at a glance

Backend A · Security hardening	20% done	~1 week
Backend B · Integration completeness	40% done	~3–4 days
Backend C · Deploy & operate	Not started	~1 week
Frontend · Real product + real data	25% done	~3–4 weeks
Models · The Brain merge	Seam ready	~1–2 wks (blocked on training)
Scale · Channels & hardening	Not started	~2–3 weeks

Backend A+B+C and the Frontend run in parallel; the Model merge waits on the trained adapters but drops in behind an existing seam.

3. Track detail — done vs left

Backend A · Security hardening (~1 week)

Close the audit's CRITICAL/HIGH safety items so nothing fails open.

Done

check CORS for the browser SPA
check Provider-agnostic auth + Clerk wired (string subject, migration)
check Resilient startup (analysis-DB hiccup no longer crashes the API)
check Clerk signup webhook (Svix-verified, idempotent)
Left

- Fail-closed startup guard: refuse to boot in prod without crypto keys; publish guard never skips when the signing key is blank.
- DPDP erasure completeness: add signals / comments / knowledge tables; verify cascade deletes in tests.
- Audit-purge vs hash-chain: redact-in-place instead of hard-delete so the chain still verifies.
- Content guardrails (R-2/R-3): pre-LLM input filter + post-gen brand-safety / toxicity / do-not-claim check.
- Cost ceiling: per-run + daily █ cap with a circuit-breaker; fix █0-cost for non-Groq models.

Backend B · Integration completeness (~3–4 days)

Done

check GET /me (who-am-I)
check Intake endpoints wired end-to-end from the browser
Left

- HTTPBearer security scheme so /docs + generated clients show the auth requirement.
- GET /founders/{id}/runs — list a founder's run history.
- Machine-readable run/item status enums + documented state machine for the UI.
- Run progress: a clean polling contract (or SSE) so the dashboard can follow a multi-minute run.

Backend C · Deploy & operate (~1 week)

Turn the doc-only production architecture into runnable, recoverable artifacts. Some steps need accounts (flagged).

- Containerize: Dockerfile + .dockerignore.
- Deploy manifest (render.yaml) — web + worker; CI runs migrations then deploys with rollback. [needs Render]
- Background worker (ARQ + Redis): scheduled signal/comment pulls, stale-approval expiry, run auto-resume. [needs Redis/Upstash]
- Observability: Sentry + structured logs + /metrics; alerts on guardrail/publish failures. [needs Sentry]
- Backups: scripted pg_dump + restore runbook; startup migration-version check.
- CI hardening: Postgres service so RLS/tenant tests run; coverage gate.

Frontend · Real product + real data (~3–4 weeks) — the biggest piece

Done

check Clerk login + auth guard on the app
check Cowork onboarding form wired to the real backend (saves the founder profile)
Left

- Cowork workspace + Brain read the real saved profile (not mock beliefs/docs).
- Runs surface: start a run, poll status, render the Gate-1 intelligence report + generated content items.
- Gate-1 / Gate-2 approval UI inside the app (per-item approve / reject-with-note / regenerate) — the core trust feature.
- Connections: real OAuth (start → provider → callback) replacing the simulated connect.
- Decide the marketing-analytics screens (leaks/channels/funnel/weekly-report) — they have NO backend; re-scope to the real product or retire.
- Delete lib/mock once each surface is real.

Models · The Brain merge (~1–2 weeks, blocked on training)

The trained Qwen+LoRA Brain and self-hosted RAG drop in behind the existing model seam: config-driven per-role endpoints, an LLM-as-judge eval gate at 70% before flipping any seat, rented Groq demoted to fallback. No code rewrite — a configuration milestone.

Scale · Channels & hardening (~2–3 weeks)

- Redis-backed rate limiter + JWKS cache (multi-replica correct).
- OAuth token auto-refresh; live LinkedIn/X publishing transports with backoff.
- Indic / multilingual generation — the India differentiator.
- Meta Ads assisted-draft + Reels (remaining V1 channels); k6 load test; full compliance posture.

4. Going real-time: the mock-data reality

"Remove the mock data" is not one switch — each surface falls into one of three buckets. The honest map:

Sign-in / sign-up	✓ Clerk	DONE — real.
Onboarding (Cowork)	✓ intake API	DONE — saves real founder profile.
Cowork workspace + Brain	Partial	Read the real saved profile; seed beliefs from it. Runs/insights later.
Console · Approvals	✓ Gate-2	Wire to real runs + per-item approval. (needs runs live)
Console · Eagle-View / Campaigns	✓ signals/runs	Wire to real run + signal data.
Console · Comments	✓ comment agent	Wire to real comment endpoints.
Console · Brain-mini / roll-up	Partial	Derive from real profile + runs.
Old /dashboard (leaks/channels/funnel)	✗ none	DECISION: re-scope to the real product or retire.

Key point: the new /console dashboard maps cleanly to real endpoints; the old marketing dashboard does not. So "go real-time" = wire /console to the real run/approval/signal pipeline, and make a call on the old screens.

5. How much more to a production-ready site

Now → real-time V1	De-mock /console onto real runs/gates/profile; finish Backend A+B.	~2–3 weeks
Deployable	Backend C: container, deploy, worker, observability, backups.	~1 week
Founder-usable V1	Full intake → run → approve → export, real data, deployed.	= ~5–7 wks total
Intelligent	Connect the trained Brain + RAG behind the seam.	+1–2 wks (on models)
Scale-ready	Redis, live publishing, Indic, more channels, load test, compliance.	+2–3 weeks

Bottom line: a deployed, real-data, founder-usable V1 (everything except the trained models, which connect via the seam) is roughly 5–7 weeks of focused work. The critical path is the frontend rebuild on the real product — the backend is the strong, mostly-finished half.

6. Immediate next — the real-time sequence

Recommended order to remove mock and go real, highest-value first:

- 1 Cowork reads the REAL founder — workspace + Brain seeded from the saved profile (data already exists in Supabase).
- 2 Finish Backend B (list-runs, status enums, HTTPBearer) so the dashboard can talk to runs.
- 3 Wire /console Approvals to a real run + Gate-2 (start a run → report → per-item approve).
- 4 Wire Eagle-View / Campaigns / Comments to real run + signal + comment data.
- 5 Decide & handle the old marketing dashboard (re-scope or retire).
- 6 Delete lib/mock surface by surface as each goes real.

MRK18 — Production Roadmap. Living document; figures assume a two-founder pace.